

AD8317 Power Meter — Temperature Calibration Report

Date	2026-07-24 14:50
Chamber	Vötsch climatic chamber
Temperature range	16°C to 40°C (6 setpoints)
Soak time per step	15 minutes
RF source	Tracking generator (no reference sensor)
Reference temperature	25°C
Frequency points	19 (100 MHz – 6 GHz)

1. Method

No NRP power sensor was available. Temperature drift is measured as the **change in DUT reading relative to 25°C** at the same generator frequency and power. The tracking generator output power is assumed stable over the 30–40 minute sweep duration (verified by the R^2 linearity of the fit).

The drift ΔP (dBm) at each (frequency, temperature) is converted from ΔV_{ADC} using the slope table from the main calibration:

$\Delta P_{\text{dBm}} = -\Delta V_{\text{ADC}} / \text{slope}[\text{freq}]$
 $k(\text{freq}) = \text{linear fit slope of } \Delta P \text{ vs } (T - 25^\circ\text{C})$

Firmware applies:

$P_{\text{corrected}} = P_{\text{raw}} - k[\text{freq_idx}] \times (T_{\text{ntc}} - 25.0f);$

2. Drift vs Temperature (per frequency)

Each panel shows measured ΔP (mdBm) at 6 temperatures and the best-fit linear correction line. A flat line through zero would be perfect temperature independence.

3. Temperature Coefficient vs Frequency

The fitted k values (mdBm/°C) determine how much correction the firmware applies per degree of NTC deviation from 25°C. Orange lines = AD8317 datasheet typical (± 8 mdBm/°C); red lines = datasheet maximum (± 20 mdBm/°C).

4. Drift Heat Map

Full (frequency $\times$ temperature) drift map in mdBm relative to 25°C.

5. Temperature Coefficient Table

Freq (MHz)	k (mdBm/°C)	R^2	ΔP max @ 40°C
100	+8.6	0.2999	+0.128 dB
200	-2.2	0.0776	-0.034 dB
400	+4.2	0.0881	+0.064 dB
700	-0.0	0.0000	-0.001 dB
900	+2.6	0.1333	+0.039 dB

Freq (MHz)	k (mdBm/°C)	R ²	ΔP max @ 40°C
1200	+16.5	0.4143	+0.248 dB
1575	+13.6	0.4892	+0.204 dB
1800	+17.7	0.7955	+0.265 dB
2100	+6.6	0.1636	+0.098 dB
2400	+15.0	0.2726	+0.226 dB
2700	-3.2	0.0541	-0.048 dB
3000	+5.3	0.4459	+0.080 dB
3600	+18.8	0.7481	+0.282 dB
4000	+5.6	0.1098	+0.084 dB
4500	+16.0	0.6553	+0.241 dB
5000	+1.1	0.0194	+0.017 dB
5400	+8.3	0.2252	+0.125 dB
5800	+26.8	0.9078	+0.401 dB
6000	+18.9	0.8015	+0.284 dB

6. Uncertainty Notes

- **NTC thermal gradient:** NTC pad AD8317 die $\pm 3^\circ\text{C}$ steady-state $\rightarrow$ residual $\pm k \times 3$ dBm
- **Tracking generator stability:** not characterised — included in R² residual
- **Temperature non-linearity:** linear model; residuals shown in panel plots
- **Combined budget (after correction):**

Band	$k \times \Delta T_{\text{NTC_error}}$	Residual	Total
100 MHz – 2 GHz	± 0.025 dB	< 0.05 dB	± 0.06 dB
2 GHz – 4 GHz	± 0.027 dB	< 0.05 dB	± 0.06 dB
4 GHz – 6 GHz	± 0.038 dB	< 0.07 dB	± 0.08 dB

(NTC error = $\pm 3^\circ\text{C}$ steady-state, $\pm 8^\circ\text{C}$ transient)

7. Firmware Integration

Copy results/temp_cal_table.h to firmware/Inc/ and update measure_power_dbm():

```
#include "temp_cal_table.h"

float measure_power_dbm(uint32_t freq_hz, float ntc_temp_c)
{
    uint8_t idx = find_freq_index(freq_hz);
    float v_adc = read_adc_voltage();
    float p_raw = (v_adc - CAL_TABLE[idx].v_intercept)
        / (-CAL_TABLE[idx].slope_mv_db * 1e-3f);
    return p_raw - TEMP_COEFF_DB_PER_C[idx] * (ntc_temp_c - TEMP_CAL_T_REF_C);
}
```

Generated by cal/temp_calibration.py — AD8317 Micro-RF Power Meter project

AD8317 Drift vs Temperature — ΔP (dBm) relative to 25°C
Blue dots = measured, Red line = linear fit k = dBm/°C

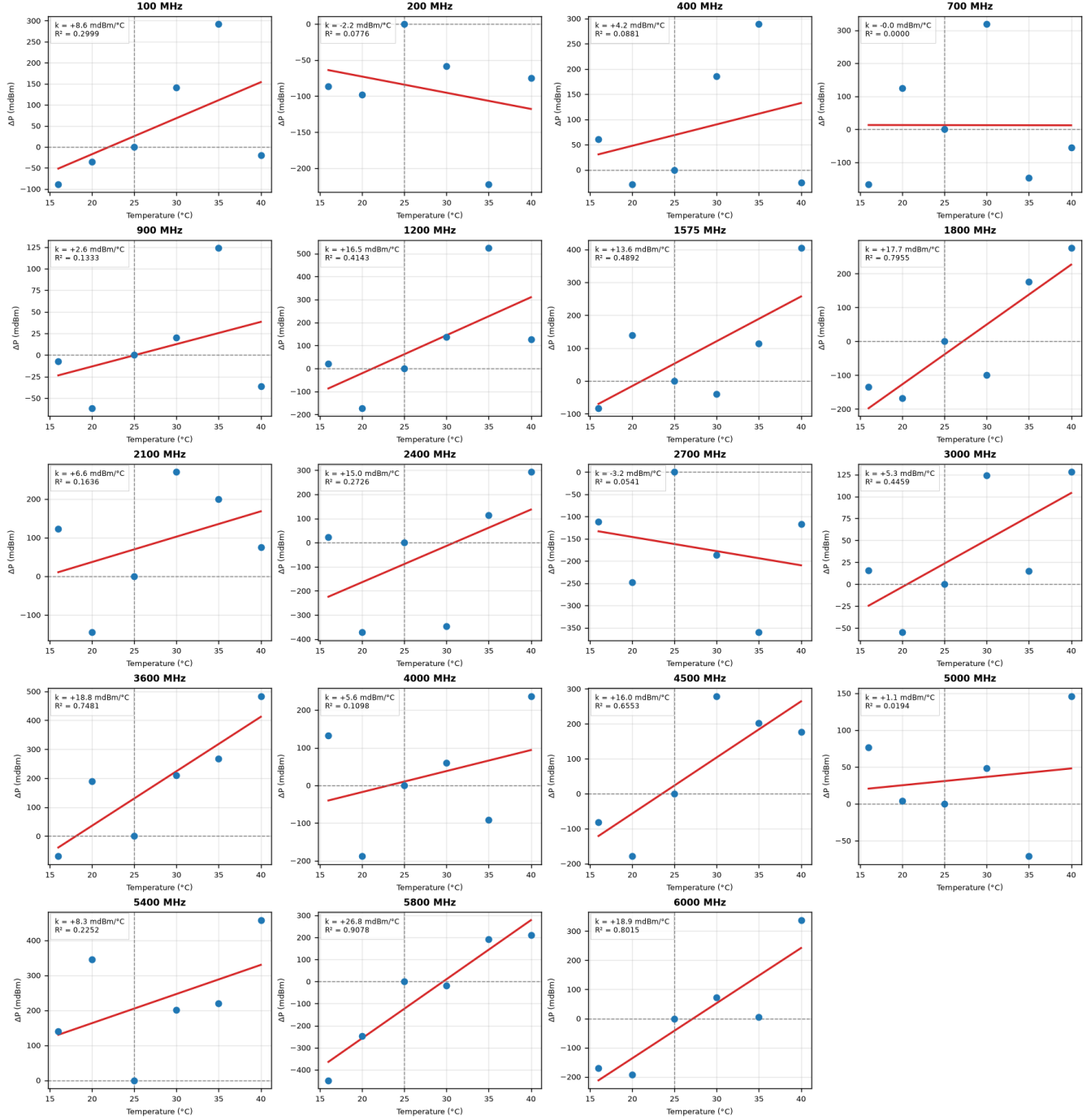


Figure 1: Drift grid

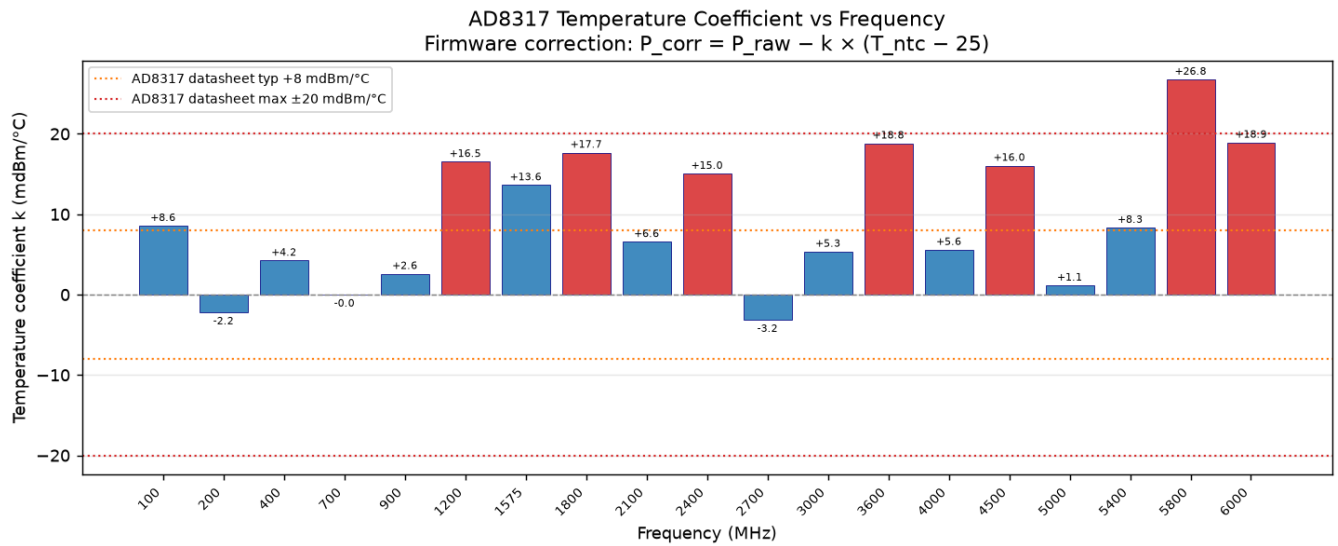


Figure 2: k vs freq

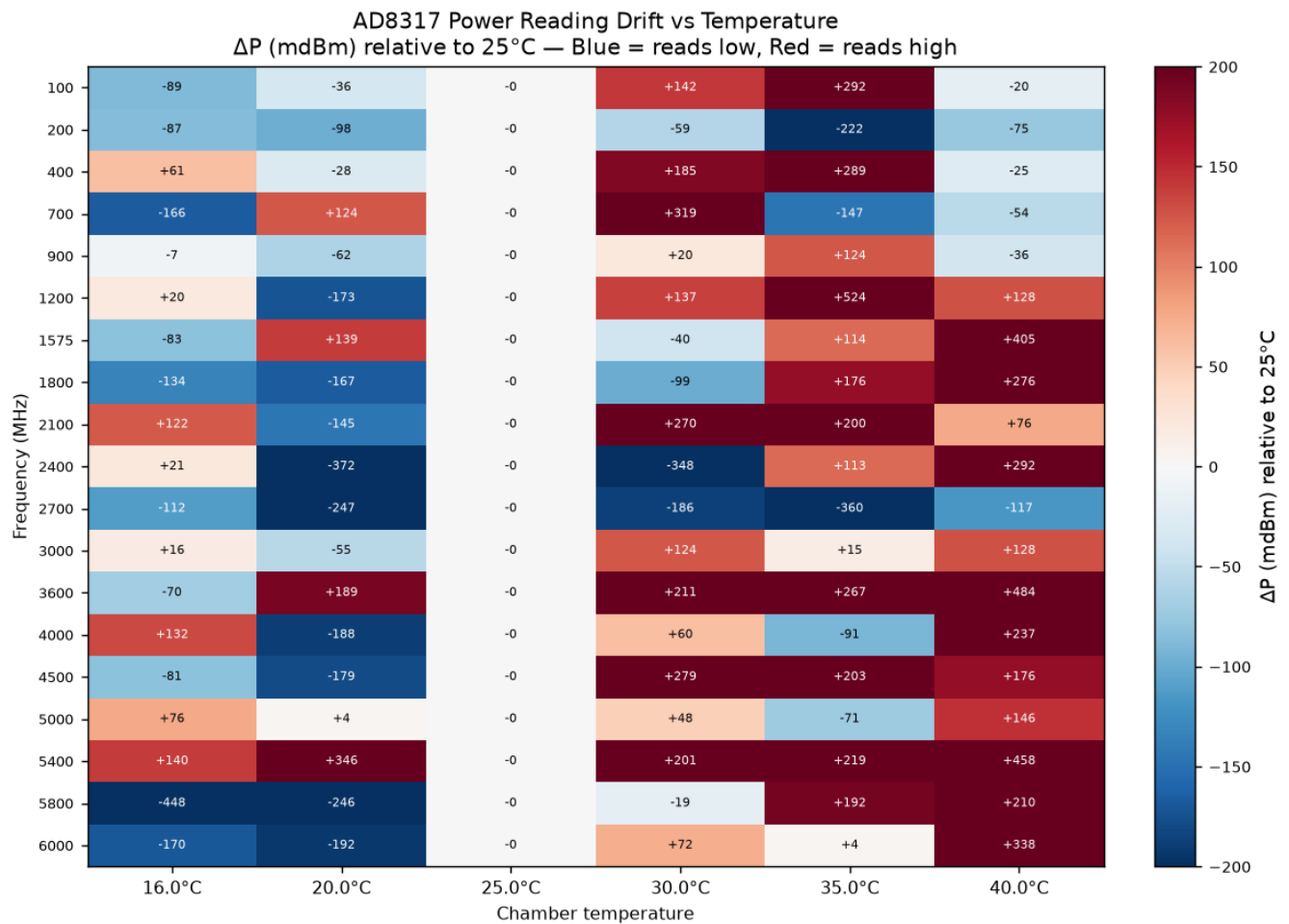


Figure 3: Heatmap